

PORTFOLIO PROJECT

Aegis Sentinel

A malware, phishing and cleanup tool for Windows.

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THE STORY

From a Python script to a Windows app

Before

Skull Sentinel, my Python project with a web style window and packaged executables.

Now

Aegis Sentinel, one C# program. No Python, no installer, built with the compiler already inside Windows.

The Aegis family

Same look and feel as Aegis DiskGuard, so my tools feel like one product.

WHAT IT DOES

Eight tools in one window

Malware Scan
Finds known bad files and disguised programs.

URL Check
Scores a link before you click it.

Message Check
Spots scam wording and risky links.

Device Tune-Up
CCleaner style junk file cleaner.

Quarantine
Isolated files you can restore any time.

Startup Manager
Turn startup programs off and on.

Activity Log
A history of everything the app did.

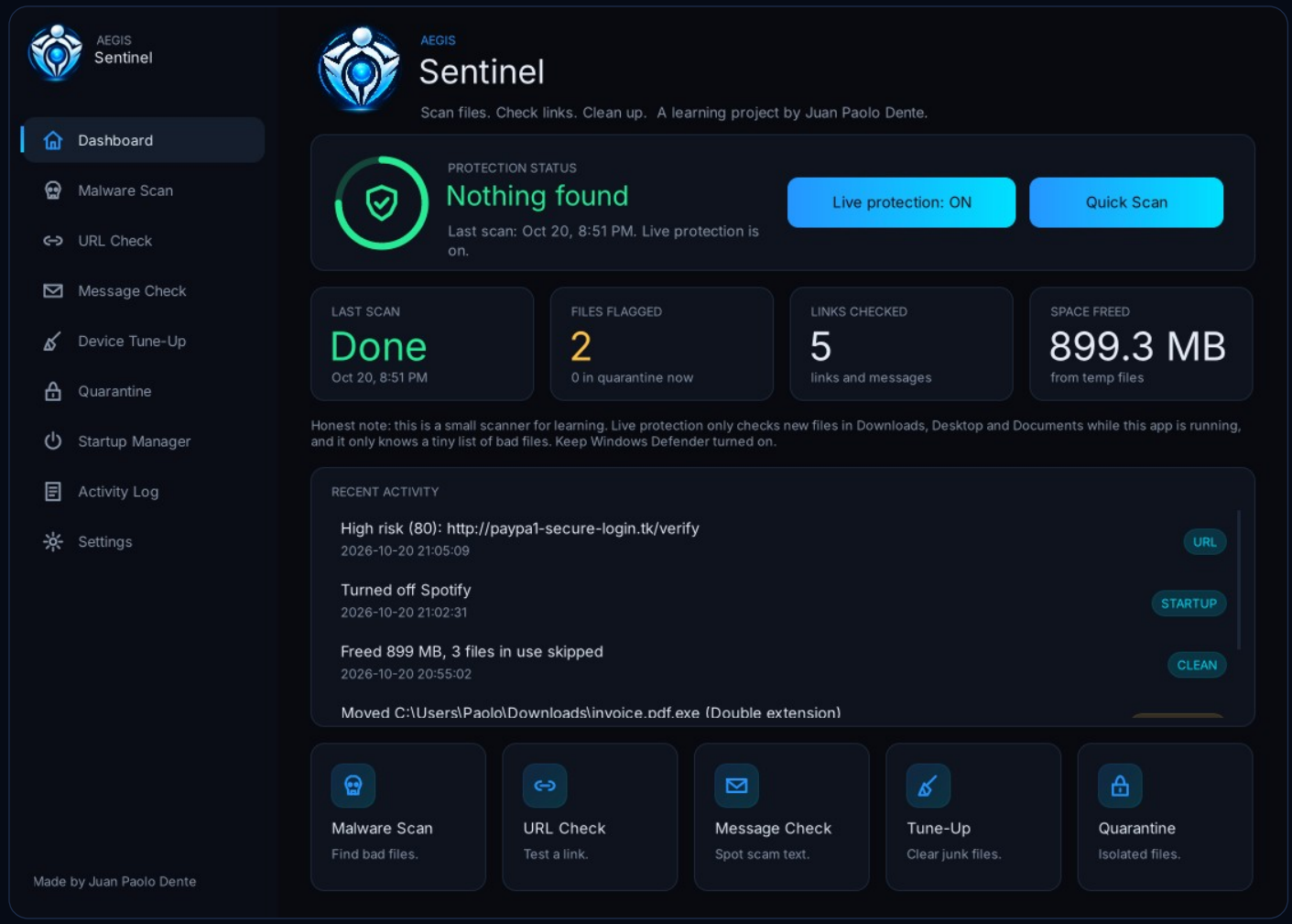
Live protection
Checks new files as they arrive.

TOUR

Dashboard

One screen shows protection status, your stats and recent activity.

- Quick Scan button
- Live protection switch
- Last scan, files flagged, links checked, space freed

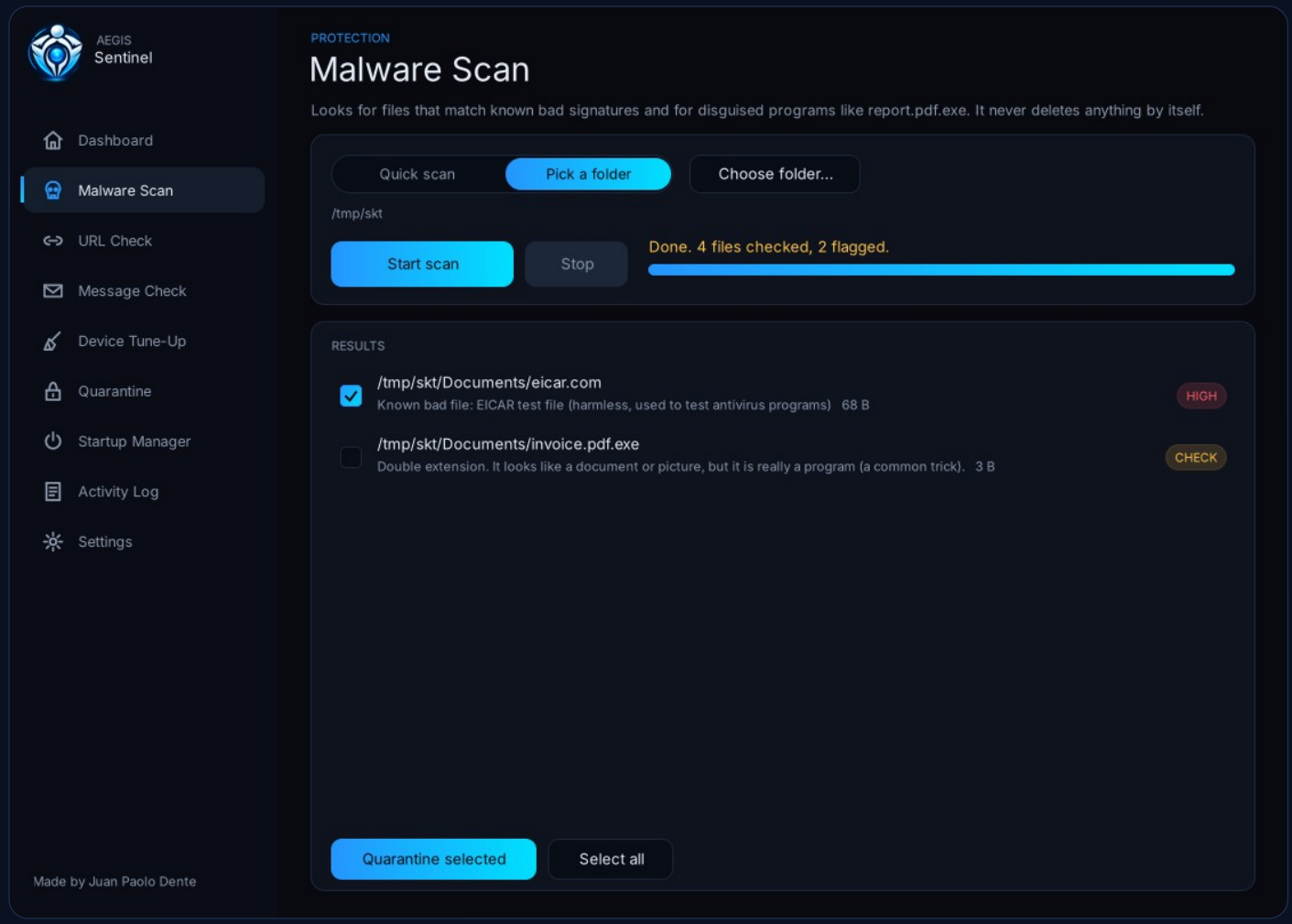


TOUR

Malware Scan

Two checks on every file, and nothing is ever deleted on its own.

- SHA-256 fingerprint compared with known bad files
- Catches tricks like invoice.pdf.exe
- Flagged files go to Quarantine

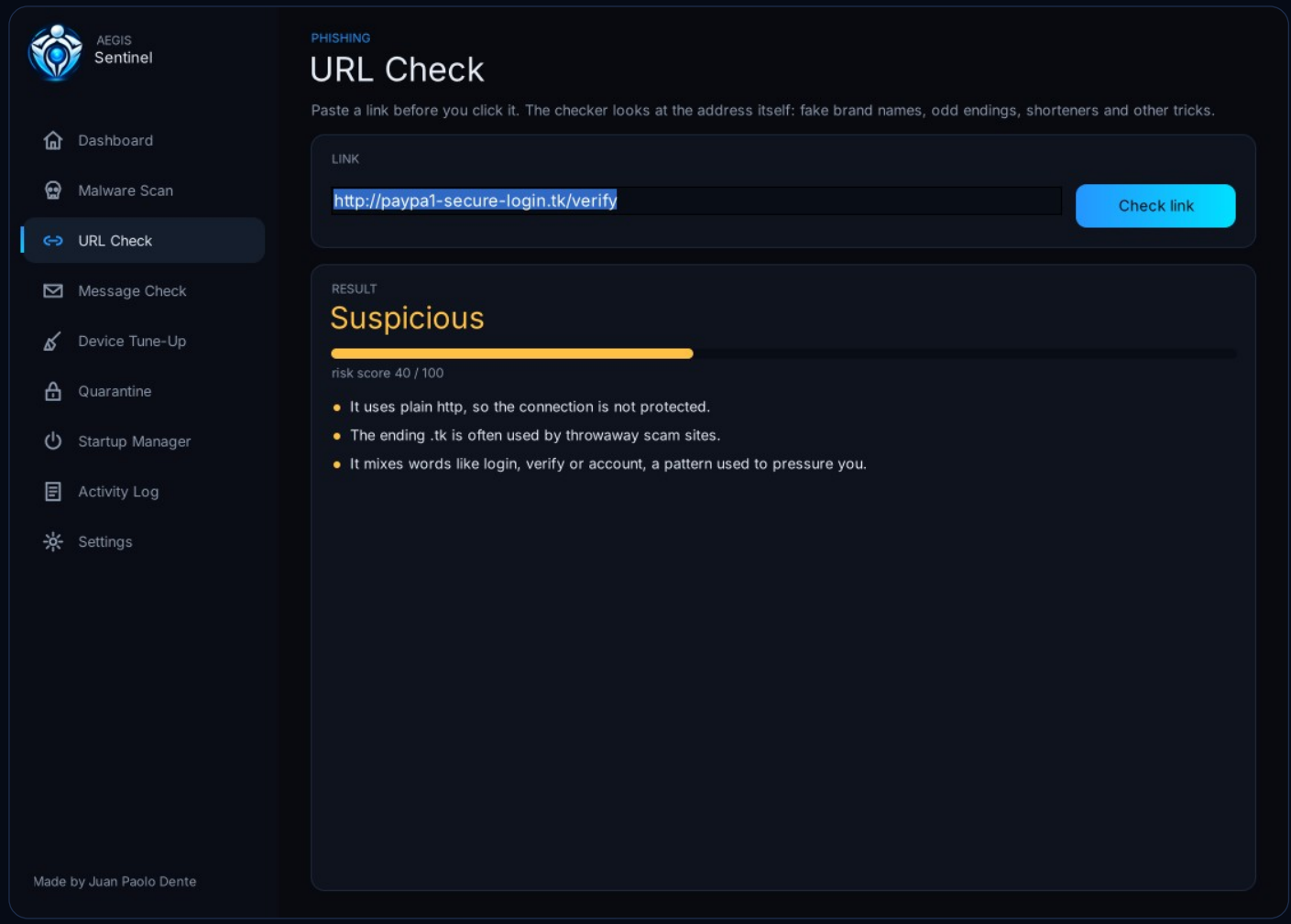


TOUR

URL Check

Paste a link and get a risk score with plain reasons.

- Fake brand names and look-alike spelling
- Risky endings, raw IP addresses, link shorteners
- Words like login and verify in odd places

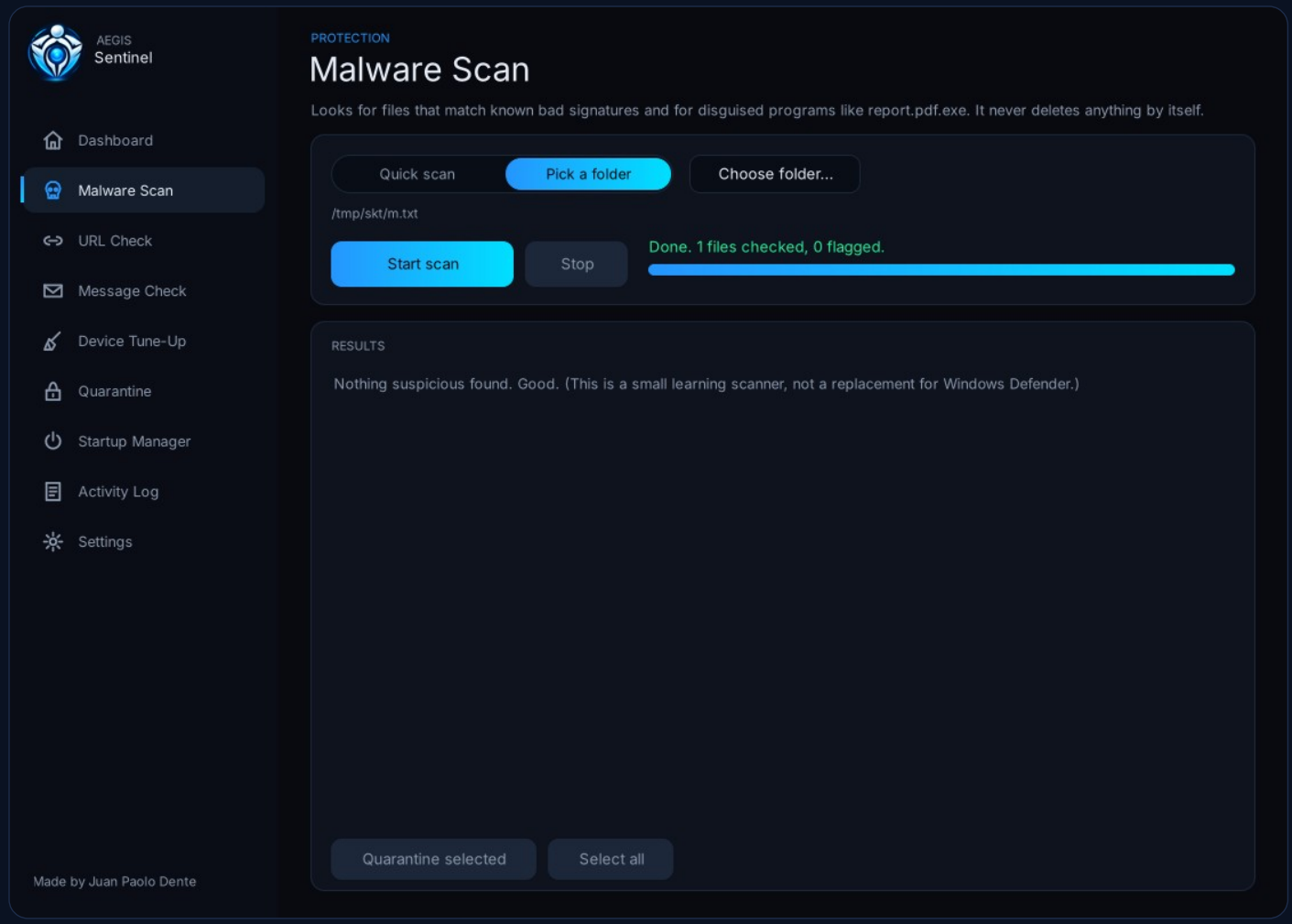


TOUR

Message Check

Paste an email or text and see why it looks like a scam.

- Pressure words and deadlines
- Requests for passwords or OTP codes
- Every link inside is scored too

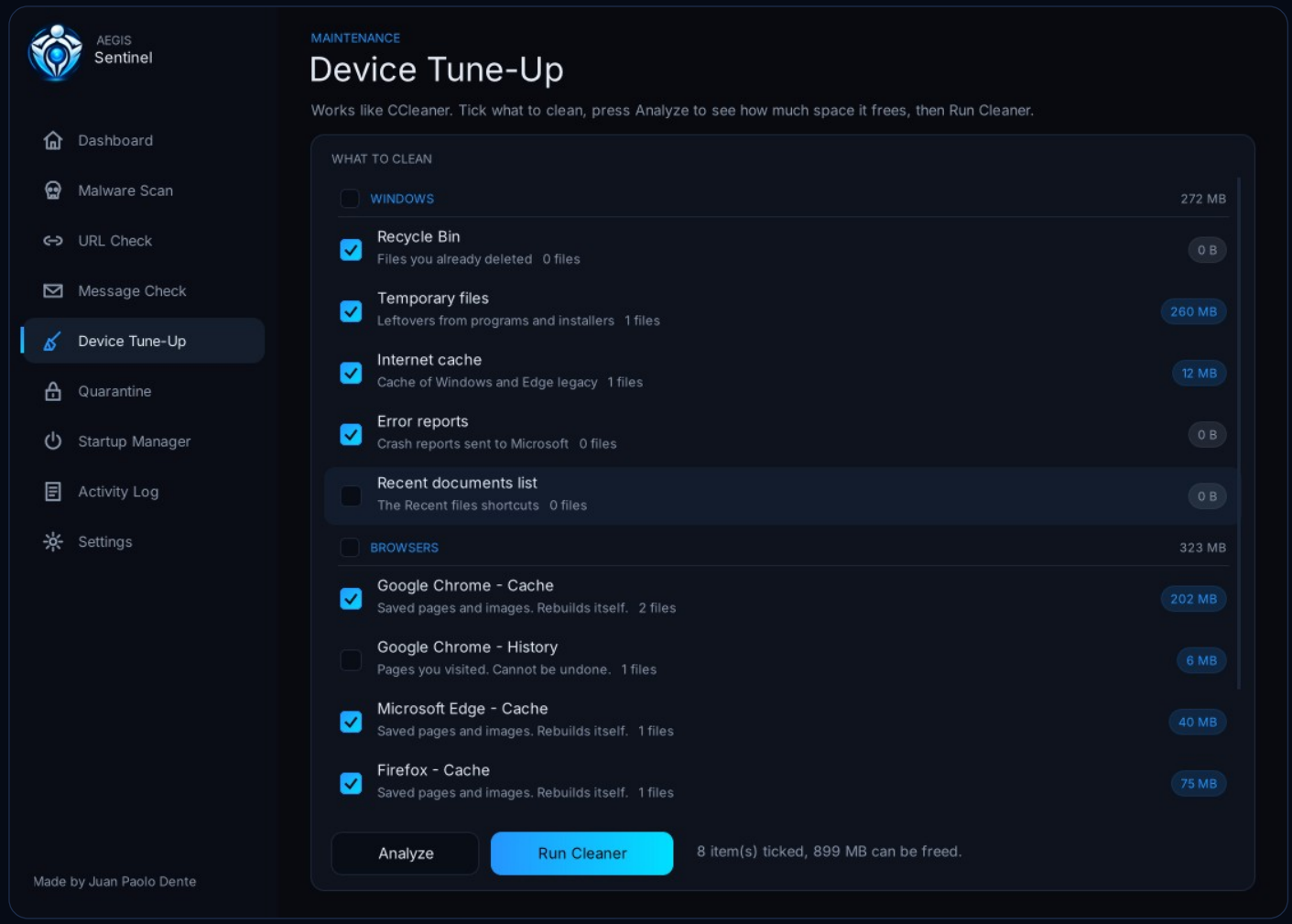


TOUR

Device Tune-Up

A CCleaner style cleaner. Analyze first, then Run Cleaner.

- Groups: Windows, Browsers, Applications
- History and cookies start unticked
- Warns when a browser is still open



HOW IT WORKS

Live protection in five steps

1

A new file lands

In Downloads, Desktop or Documents.

2

It waits

Until the download or copy is finished.

3

It checks

Fingerprint and double extension test.

4

It alerts

A popup from the tray icon.

5

You decide

Quarantine it, or leave it alone.

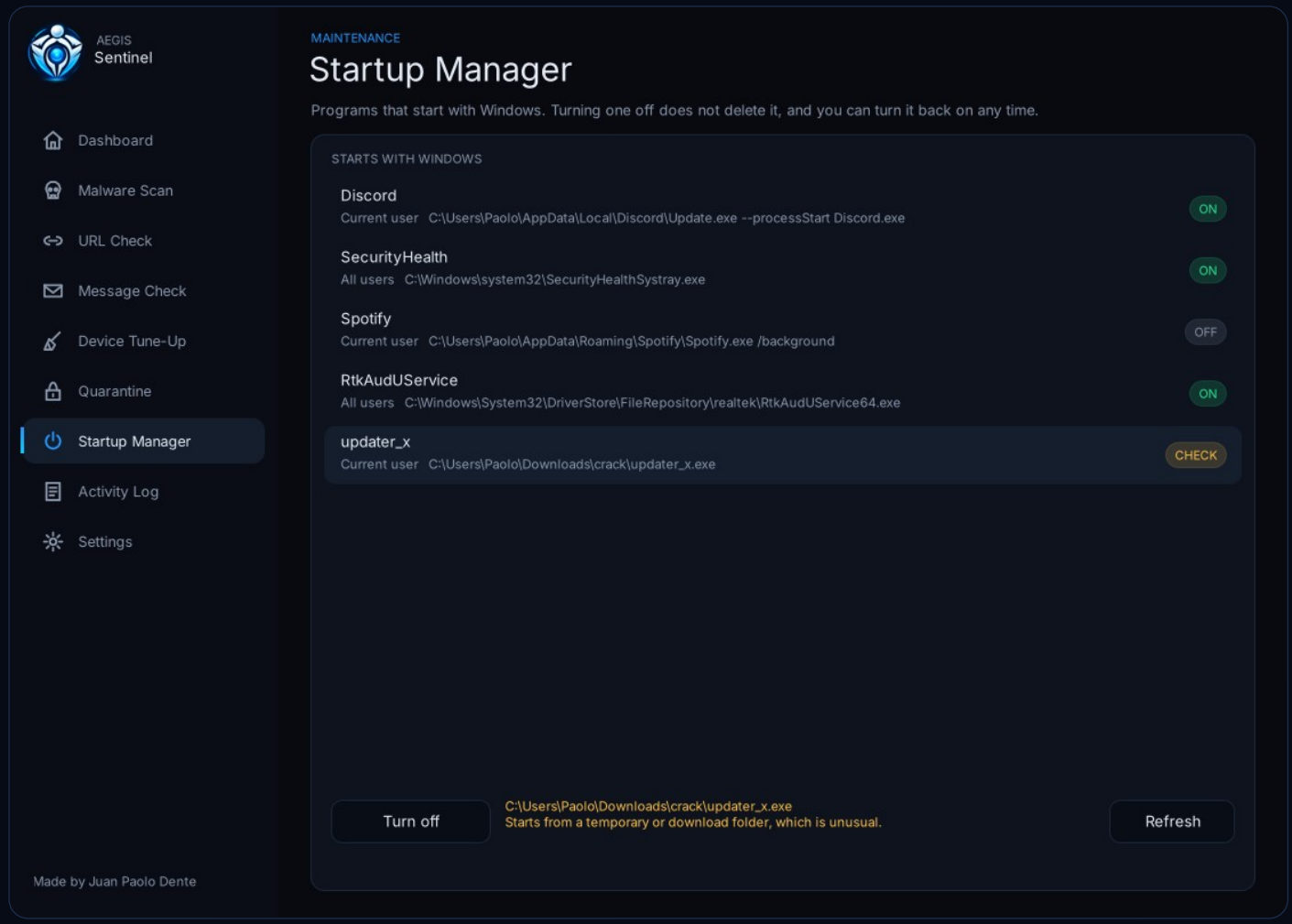
It can also start with Windows, hide in the tray and quarantine known bad files automatically if you turn that on.

TOUR

Startup Manager

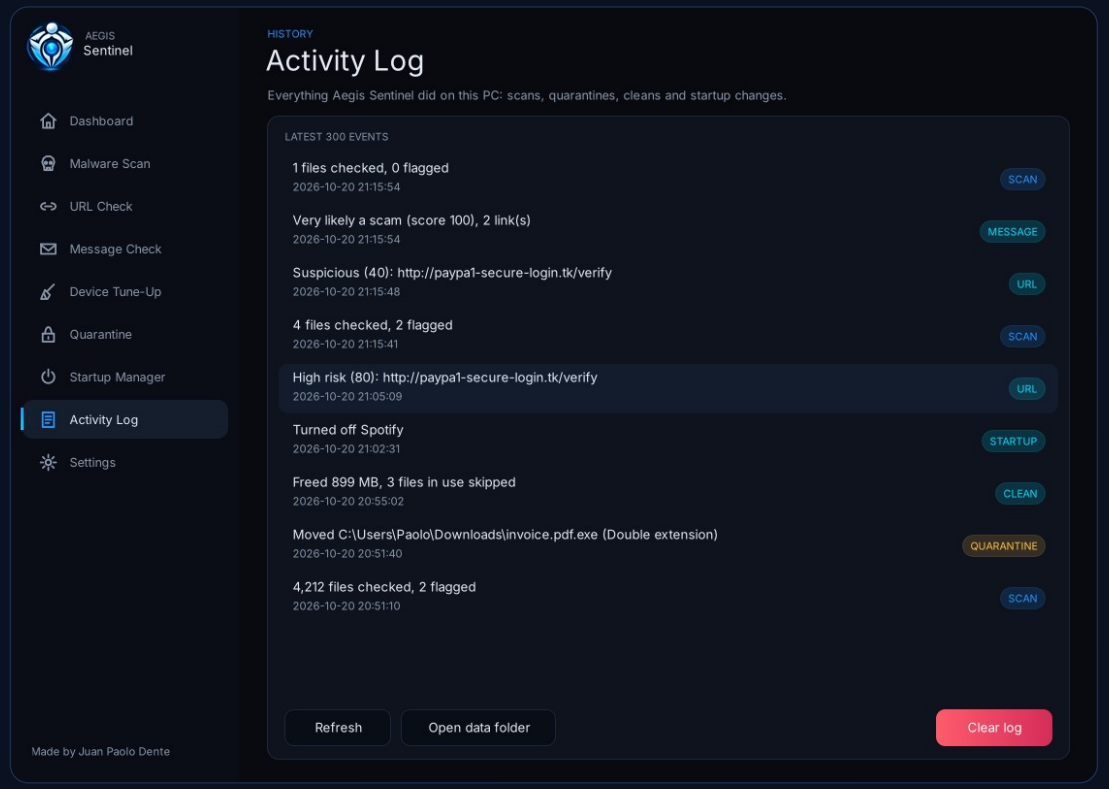
See what starts with Windows and switch it off without deleting anything.

- Uses the same switch as Task Manager
- Odd entries are marked CHECK
- Every change goes in the Activity Log

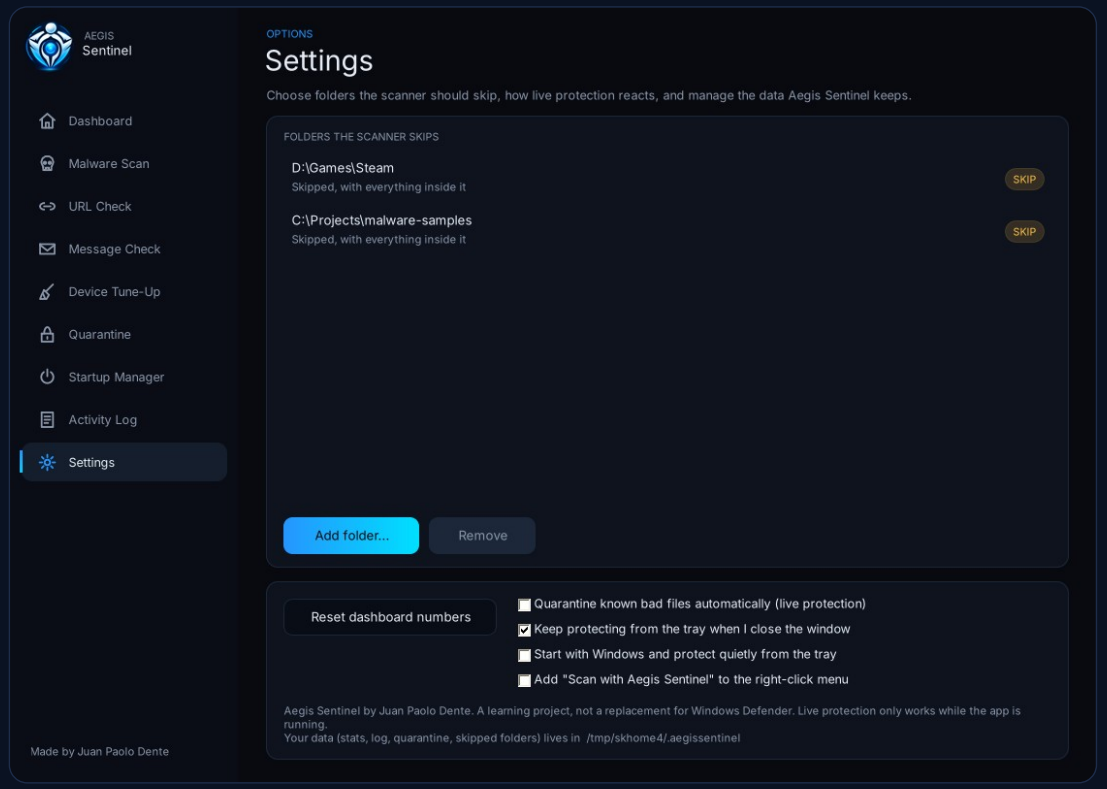


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Activity Log and Settings



A clear history of scans, cleans and quarantines.



Skip folders, start with Windows, add a right-click scan.

HOW IT IS BUILT

One program, five code files

The engine

Engine.cs and Extra.cs hold the scanning, cleaning, quarantine, startup, log and live protection code. No screens in them.

The look

UI.cs and Pages.cs draw the dark theme with custom controls, so it matches Aegis DiskGuard.

The window

Program.cs holds the dashboard, the tray icon and the main window.

C# with Windows Forms. A single build.bat uses the compiler already in Windows, so there is nothing to install.

TESTING AND LIMITS

What works, and what it is not

Tested on my PC

- Live protection flagged a fake invoice.pdf.exe with a popup
- The tray icon turns on and off with the switch
- Startup Manager lists my real startup programs
- It starts with Windows and stays in the tray

Honest limits

- A learning project, not a replacement for Windows Defender
- Only a tiny list of known bad files
- Live protection watches three folders, while the app runs
- Link and message checks are rules of thumb

WHAT IS NEXT

Where Aegis Sentinel goes from here

A bigger bad-file list

Pull fingerprints from a free source so more threats are known.

An Aegis launcher

One small window that opens DiskGuard, Sentinel and AEGISWFP.

Scheduled scans

A quick scan every day with a notification.

Code signing

So Windows SmartScreen stops warning when the app is shared.



Thank you. Built by Juan Paolo Dente.